

NOAM J. GAL – CURRICULUM VITAE

PERSONAL DETAILS

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Web Profiles: [GitHub](#), [Google Scholar](#), [LinkedIn](#)

Date and Place of Birth: January 31, 2001; Brooklyn, NY, USA

Citizenships: Israel, United States, Canada

EDUCATION

HEBREW UNIVERSITY OF JERUSALEM

Jerusalem, IL

Ph.D. Candidate, Stage A | Geoinformatics Track

Oct. 2025- Present

Topic: Geospatial Artificial Intelligence (GeoAI)

Supervisor: Professor Amit Birenboim | Dept. of Spatial Science

M.A., Smart Cities and Urban Informatics

July, 2025

GPA: 95.6 / 100

Summa Cum Laude

Thesis topic: Fragmentation of Digital-Physical Behavior

Supervisor: Professor Amit Birenboim

CITY UNIVERSITY OF NEW YORK | MACAULAY HONORS COLLEGE

New York, NY

B.Sc., Mathematics, Urban Economics

June, 2022

GPA: 3.98 / 4.0

Summa Cum Laude

PUBLICATIONS

Gal, N. J., Hanny, D., Janowicz, K., Resch, B., & Birenboim, A. GeoAI as the new frontier in geography: A review of methodologies, definitions and trends.

(Submitted to *International Journal of Geographic Information Systems*, 2026)

- Accepted to present at the 1st International Conference on Geospatial Artificial Intelligence, Ghent, Belgium in June, 2026
- Presented at the Annual Conference of the Israeli Association for Regional Science 2026, Be'er Sheva, Israel

Gal, N. J., Wang, A. L. M., Franco, A., Chaix, B., & Birenboim, A. (2026). Mental health in a digitally fragmented world: Exploring negative and positive associations across spatial contexts and gender. *Social Science & Medicine*, 401, 119337.

<https://doi.org/10.1016/j.socscimed.2026.119337>

- Awarded Best Oral Presentation the GeoHB2025 Conference on Geospatial Computing in Human Behavior, Shanghai, China
- Presented at the 6th Israeli Smart Transportation Research Center Students Conference, Haifa, Israel

Gal, N. J., Nikitin, A., Drogochinsky, M., Cohen, Y., Batat, M., Kaufmann, T., & Noyman, A. The right to shelter: A planning support system for protecting Negev Bedouin communities. (Submitted to *She Ji*, 2026)

- Tool results were presented to the Knesset State Audit Committee, Jerusalem, Israel on January 20, 2026 at a special meeting on the gaps in shelter coverage for Negev Bedouin

Wang, L. M., Montanari, A., **Gal, N. J.**, Chaix, B., & Birenboim, A. Temporal patterns of smartphone-mediated digital engagement and mental health symptomatology. (Submitted to *Computers in Human Behavior Reports*, 2025)

Wang, L. M., Montanari, A., Moser, M. K., **Gal, N. J.**, Chaix, B., Resch, B., & Birenboim, A. Moments of stress in the city: A novel ecological approach to monitoring urban and digital stressors. (Submitted to *Cities*, 2025)

Alfonso, J., Baptista, R., Bhakta, A., **Gal, N. J.**, Hou, A., Lyubimova, V., Pocklington, D., Sajonz, J., Trigila, G., Tsai, R. (2023). "A generative flow model for conditional sampling via optimal transport." <https://arxiv.org/abs/2307.04102>

- Underwent peer review, was accepted, and presented as a poster at the high-impact NeurIPS 2023 conference in the Optimal Transport and Machine Learning Workshop, New Orleans, Louisiana, USA

AWARDS AND HONORS

Leventer Urban Planning Prize (led data science for winning project, awarded to NUR by Tel Aviv University), 2025

Kate Klausner Scholarship, (Hebrew University of Jerusalem), 2025-2030

Prof. Amiran Scholarship, (Spatial Science Dept., Hebrew University), 2025-2026

Best Oral Presentation Award, (GeoHB2025 Conference), 2025

Conference Grant, (Israeli Smart Transportation Research Center), 2025

Fulbright Master's Degree Fellow, (U.S. Department of State), 2023-2024

Provost Award, (Macaulay Honors College), 2022

Portz Interdisciplinary Fellow, (National Collegiate Honors Council), 2021-2022

Community Semester Fellow, (City University of New York), 2021

Thomas W. Smith Fellow, (City University of New York), 2020-2021

Dean's List, (City University of New York), 2018-2022

Macaulay Honors Scholarship, (Macaulay Honors College), 2018-2022

RESEARCH EXPERIENCE

URBAN VITALITY LAB

Jerusalem, IL

Graduate Research Scientist

Feb. 2024 – June 2025

- Analyzed and preprocessed geospatial, physiological, and digital environment data for several lab studies resulting in two submitted articles and one published peer-review article
- Designed unsupervised learning methods and mixed effects models for detecting patterns in movement patterns in the urban environments with significant effects on anxiety levels

POLYMATH REU

Virtual

Machine Learning Research Assistant

June 2022 – May 2023

- Coauthored research article presented at NeurIPS 2023 on solving the optimal transport problem of mapping probability densities from independent samples to joint distributions
- Built an Input Convex Neural Network using a custom loss function to find optimal maps that established baseline replication for the current state of the art

TEACHING EXPERIENCE

HEBREW UNIVERSITY OF JERUSALEM

Jerusalem, IL

Graduate Teaching Assistant, Department of Spatial Science

(Oct. 2024 – Present)

1. Foundations of Spatial Statistics (Course: 40733)
 - Lectured, prepared exercises, graded exercises
2. Researching Smart Cities (Course: 01678)
 - Lectured, prepared exercises, graded exercises

PROFESSIONAL EXPERIENCE

NEGEV URBAN RESEARCH (MIT CITY SCIENCE LAB)

Beer Sheva, Israel

Lead Data Scientist

June 2025 – Present

- Leads a cross-functional team of data analysts, geospatial designers, and developers in several end-to-end data science projects from data collection/acquisition through full data analysis and cleaning pipeline to deliver to clients and public as data-driven web applications
- Owns and maintains full stack application architecture urban analytics platform built upon MIT Cityscope that is used for planning the Be'er Sheva Innovation District
- Developed community engagement interface to collect feedback on plans to construct memorial sites and tourist infrastructure in the Gaza Envelope

Geospatial Data Scientist

July 2024 – June 2025

- Led development of a web application to locate optimal locations for bomb shelters in unrecognized Bedouin villages that was presented in the Knesset and published in Ynet
- Designed ETL pipelines and built urban simulation models in Python to optimize infrastructure planning for 50,000+ users of the Beer Sheva Innovation District

- Built and deployed a web platform for 3D visualizations of mobility patterns using JavaScript (React, deck.gl) frontend and Python backend

NYC MAYOR'S OFFICE

New York, NY

Data Analyst, Office of Workforce Development

Jan. 2022 – May 2022

- Created, conducted, and analyzed a survey of workforce development board directors throughout New York State that was used as evidence for a successful application to change federal spending quota
- Automated data analysis processes required for regulatory reports in Python and SQL saving hundreds of labor hours in Excel

Community Semester Fellow, Office of Workforce Development

Sept. 2021 – Jan. 2022

- Conducted fiscal planning with top-level representatives of government agencies (NYSDOL, DYCD, SBS), subcontractors, and consultants to ensure regulatory compliance with the Workforce Innovation and Opportunity Act for over \$70 million in spending
- Improved UI design, created content, and wrote newsletters for WorkingNYC— a website dedicated to workforce development opportunities driving a 200% increase in site traffic
- Volunteered for a crisis task force to manage the City's emergency response to Hurricane Ida, coordinating hundreds of visits by city workers to repair damage and remove debris from affected households

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript/TypeScript

Software: ArcGis, QGIS, Geoda, Google Cloud Platform, Google Earth Engine, Stata, Docker, Postgres, Supabase, Git, Github Actions, MS Office

Core Competencies: Data Science (Geospatial Domain Expert), ML Engineering, Spatial Statistics, Applied Math, Data Visualization, ETL Pipelines

REFERENCES

- Dr. Amit Birenboim; Assoc. Professor; Hebrew University of Jerusalem;
amit.birenboim@mail.huji.ac.il
- Yonatan Cohen; Director of Technologies; Negev Urban Research;
yonatag@post.bgu.ac.il
- Chris Neale; Deputy Director; NYC Mayor's Office of Workforce Development;
CNeale@talent.nyc.gov